

LT3 Device Specification

Badge Tag with a Rechargeable Battery

Overview

The LT3 is a **rechargeable LiPo battery powered Tag** device to be used as a wearable **Tag** where tracking or navigation of persons or assets within a Ultra-Wideband / Bluetooth Real-Time Location System (RTLS) is required. Practical and robust design allows easy tracking of your personnel using lanyards or clip-ons in **industrial environments**.



Figure 1: Front and back views of the LT3 device

Key Features

- Supports both TDoA (Time Difference of Arrival) as well as TWR (Two-Way Ranging) location measurement techniques. Location accuracy sub-0.5m or better.
- Supports both UWB channel 5 (6.5 GHz) and channel 9 (8 GHz) with UWB RF compliance compatible with FCC, Japan's ARIB and ETSI of the CE.
- With Low Noise Amplifier, the device provides superior UWB range coverage.
- Integrated Motion sensor: 3-axis accelerometer.
- Three multi-function buttons.
- RGB front and 2 Green side LEDs for status indication.
- Buzzer for alarm functions.
- Firmware update over Bluetooth and USB.
- Holder for front sticker. Holder for back sticker or card holder.

- Free software configuration and visualization tools (for Android, iOS, Windows, macOS and Linux).
- An open [online documentation](#) and [community forum](#).

Electrical Parameters

Parameter	Value
Certification	FCC, CE, ARIB (Japan)
USB (power and data)	5VDC 500mA max
Battery	LiPo 3.7VDC 1000mAh
Battery lifetime	UL-TDOA: 12 months to 5 years depending on the update rate TWR: 3 months to 3 years depending on the update rate
Operating temperature (without battery) ¹	-40°C - +85°C
Operating temperature (with battery) ²	-20°C - +45°C
Operating temperature (charging) ³	0°C ~ +40°C
UWB supported channels	Channel 5 (6240–6739.2 MHz, CF=6489.6 MHz) Channel 9 (7738–8237.2 MHz, CF=7987.2 MHz, FiRa compatible)
UWB transmit powers	FCC/ARIB/ETSI: -41.3 dBm/MHz max
Bluetooth	2.4 GHz, +8dB

Mechanical Parameters

Parameter	Value
Size	W=75 x H=95 x D=15 mm
Weight	With battery: 72g
Material	ABS

Color	White
IP rating	IP67
Mounting	Lanyards or clip-ons

Safety Instructions

- Do not expose devices to water or moisture while in operation.
- Do not expose devices to excessive heat from any source. The product is designed for reliable operation at industrial temperature¹. If the product is equipped with a battery, it can be at normal operating temperature². The temperature³, while the battery is charging, is internally protected (if the temperature is outside of the range, the battery charging will not start or will be postponed).

Warnings

- This product shall be connected only to the USB with a minimum current supply of 0.5 A.
- Any external power supply used with this product shall comply with relevant regulations and standards applicable in the country of intended use.
- This product should be operated in a ventilated environment with a non-condensing environment and should not be covered when being operated.
- There are no user-serviceable parts inside the product, and opening the unit will likely damage the product and invalidate the warranty.
- The cables and connectors of all peripherals used with this product must have adequate insulation to meet relevant safety requirements.

Order Information

- Part number: LR-LT3A1
- Custom code: HS 8517.69.9000
- Packaging: 10pcs in paper box, 17 cm x 11 cm x 11 cm 770g (with one 2m USB-C cable)

About Us

At LEAPS, we specialize in delivering cutting-edge UWB-based localization technology to help businesses and people locate new values.

Our unique technology and innovative solutions created in close cooperation with leading UWB chip producer Qorvo, coupled with a comprehensive range of products, provide unmatched performance and versatility. From tags and anchors to custom-designed systems, our UWB hardware and software is built to meet the specific needs of diverse industries.

Contact Us

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